FROM ARMOUR TO AMMUNITION

HOW SMPP IS POWERING INDIA'S DEFENCE SELF-RELIANCE

India's defence sector is undergoing a transformation, driven by a renewed focus on Atmanirbhar Bharat and indigenous innovation. One company at the forefront of this shift is SMPP Limited. Known for its cutting-edge bulletproof gear, SMPP has evolved into a full-spectrum defence technology firm—now expanding into ammunition and drone systems to support India's strategic independence.

A significant milestone came recently with a major order under the Ministry of Defence's Emergency Procurement 5 (EP-5), for 27,700 Bullet Proof Jackets (BPJs) and 11,700 Advanced Ballistic Helmets. Designed to counter high-threat environments—like armour-piercing and Hard Steel Core bullets—these systems represent a critical leap in soldier protection.



SMPP hosted General Said Chanegriha, Chief of Staff of the People's National Army, Algeria

Built in India, Made for the World

With over 30 years in the field, SMPP has delivered more than 500,000 bulletproof jackets and helmets in the last few years to Indian forces, including the Army, CRPF, BSF, and Assam Rifles. Its products have also reached over 25 countries across Asia, Europe, the Americas, and the Middle East.

What sets SMPP apart is performance and indigenous technology. The BPJs delivered under EP-5 exceed global standards, capable of withstanding six armour-piercing rounds on a single plate—whereas many international norms only test one to three hits. SMPP's helmets are among the few globally that offer rifle-level protection that too from lethal hard steel core and armor piercing bullets, a game-changer in reducing battlefield head injury fatalities. The performance comes from SMPP"s 100% indigenously designed and developed technology of Boron Carbide, one of the hardest and lightest materials for bullet protection.

Innovation at the Core

SMPP's journey is deeply rooted in R&D. The company has filed 17 patents (nine granted), with technologies like CaraSTOP-B4C, an ultra-strong ceramic made from Boron Carbide. This forms the backbone of its personal protection kits and is also used in armouring aircraft.

But innovation isn't just about materials. Design matters too. SMPP's jackets feature quick-release systems, dynamic weight distribution, and 360-degree coverage—including back, groin, and enhanced neck protection. Customised helmets for Sikh soldiers—factoring in both operational comfort and cultural respect—reflect the company's user-first mindset.

Every batch of ceramic plates undergoes X-ray scanning to detect flaws, ensuring consistent quality across production. This focus on precision and reliability has made SMPP's gear both battle-ready and export-worthy.

Beyond Body Armour: A Growing Defence Portfolio

While body armour remains SMPP's flagship, the company is now making strategic inroads into new domains. It has received industrial licenses to manufacture large-caliber ammunition for tanks, artillery, and infantry weapons. A massive 800-acre facility is under construction in Himachal Pradesh, aimed at becoming one of India's largest ammunition production units.

Simultaneously, SMPP is entering the drone and counter-drone space. Some indigenous systems have already cleared preliminary trials with the Indian Army, setting the stage for future contracts in this vital area.

This diversification strengthens India's defence industrial base and reduces reliance on imports. "Our vision is to develop critical technologies that enhance India's readiness and strategic autonomy," the company notes.

Navigating India's Defence Ecosystem

Despite its achievements, SMPP acknowledges the challenges of operating in India's defence procurement landscape. Long procurement cycles and a single-buyer model pose sustainability risks—especially for MSMEs. Even after a successful contract, firms often re-enter lengthy trials for future orders, with no continuity guaranteed.

SMPP advocates greater collaboration between public sector undertakings (DPSUs) and private players. The agility and innovation of the private sector, combined with the scale and infrastructure of the public sector, can create globally competitive solutions.

Make in India, Make for the World

SMPP exemplifies the Make in India, Make for the World philosophy. Its global client list includes

SMPP ACH helmet with convertor band



Armenia, the Philippines, Romania, Finland, Colombia, and the US. Notably, it recently delivered 45,000 bulletproof plates and 60,000 helmets to Armenia, and completed a complex vehicle armouring project in Colombia.

Whether it's safeguarding soldiers, armouring aircraft, or supporting peacekeeping missions, SMPP is proving that Indian manufacturing can lead on the global stage.

Inspiring the Future

For Ashish Kansal, CEO of SMPP, the mission goes beyond business. He sees immense opportunity for India's young engineers in emerging fields like AI, robotics, quantum tech, and precision systems. But to fully harness this potential, he believes faster decision-making and streamlined funding are critical.

Quoting Robert Frost, Kansal reflects: "The woods are lonely, dark and deep. We have promises to keep, and miles to go before we sleep." His words mirror the ambition of a company—and a nation—committed to safeguarding sovereignty through homegrown strength.

SMPP Magpro vest

